

Taller

Se desea insertar en una estructura utilizando expansiones dinámicas las siguientes claves.

$$k = \{115, 96, 48, 79, 35, 26, 57, 81, 70, 64, 107, 45, 62, 98, 33, 28, 28\}$$

1. Insertar paso a paso en una estructura de cubetas con $n = 2$ y $r = 3$, densidad de ocupación para expandir $D.O. \geq 82\%$ y la función Hash $H(k) = k \bmod n$ utilizando:

a. Expansiones totales

$$H(115) = 115 \bmod 2 = 1$$

$$H(96) = 96 \bmod 2 = 0$$

$$H(48) = 48 \bmod 2 = 0$$

$$H(79) = 79 \bmod 2 = 1$$

$$H(35) = 35 \bmod 2 = 1$$

$$n = 2 \text{ y } r = 3$$

	0	1
1	96	115
2	48	79
3		35

$$D.O. = 5/6 = 83.33\% \text{ (Expansión total)}$$

$$H(115) = 115 \bmod 4 = 3$$

$$H(96) = 96 \bmod 4 = 0$$

$$H(48) = 48 \bmod 4 = 0$$

$$H(79) = 79 \bmod 4 = 3$$

$$H(35) = 35 \bmod 4 = 3$$

$$H(26) = 26 \bmod 4 = 2$$

$$H(57) = 57 \bmod 4 = 1$$

$$H(81) = 81 \bmod 4 = 1$$

$$H(70) = 70 \bmod 4 = 2$$

$$H(64) = 64 \bmod 4 = 0$$

$$n = 4 \text{ y } r = 3$$

	0	1	2	3
1	96	57	26	115
2	48	81	70	79
3	64			35

$$D.O. = 10/12 = 83.33\% \text{ (Expansión total)}$$

$H(115) = 115 \bmod 8 = 3$
 $H(96) = 96 \bmod 8 = 0$
 $H(48) = 48 \bmod 8 = 0$
 $H(79) = 79 \bmod 8 = 7$
 $H(35) = 35 \bmod 8 = 3$
 $H(26) = 26 \bmod 8 = 2$
 $H(57) = 57 \bmod 8 = 1$
 $H(81) = 81 \bmod 8 = 1$
 $H(70) = 70 \bmod 8 = 6$
 $H(64) = 64 \bmod 8 = 0$
 $H(107) = 107 \bmod 8 = 3$
 $H(45) = 45 \bmod 8 = 5$
 $H(62) = 62 \bmod 8 = 6$
 $H(98) = 98 \bmod 8 = 2$
 $H(33) = 33 \bmod 8 = 1$
 $H(28) = 28 \bmod 8 = 4$
 $H(38) = 38 \bmod 8 = 6$

$n = 8$ y $r = 3$

	0	1	2	3	4	5	6	7
1	96	57	26	115	28	45	70	79
2	48	81	98	35			62	
3	64	33		107			38	

b. Expansiones parciales

$H(115) = 115 \bmod 2 = 1$
 $H(96) = 96 \bmod 2 = 0$
 $H(48) = 48 \bmod 2 = 0$
 $H(79) = 79 \bmod 2 = 1$
 $H(35) = 35 \bmod 2 = 1$

$n = 2$ y $r = 3$

	0	1
1	96	115
2	48	79
3		35

D.O. = $5/6 = 83.33\%$ (Expansión parcial)

$H(115) = 115 \bmod 3 = 1$
 $H(96) = 96 \bmod 3 = 0$
 $H(48) = 48 \bmod 3 = 0$
 $H(79) = 79 \bmod 3 = 1$
 $H(35) = 35 \bmod 3 = 2$
 $H(26) = 26 \bmod 3 = 2$
 $H(57) = 57 \bmod 3 = 0$

$$H(81) = 81 \bmod 3 = 0$$

$$n = 3 \text{ y } r = 3$$

	0	1	2
1	96	115	35
2	48	79	26
3	57		
Col	81		

$$D.O. = 8/9 = 88.89\% \text{ (Expansión parcial)}$$

$$H(115) = 115 \bmod 4 = 3$$

$$H(96) = 96 \bmod 4 = 0$$

$$H(48) = 48 \bmod 4 = 0$$

$$H(79) = 79 \bmod 4 = 3$$

$$H(35) = 35 \bmod 4 = 3$$

$$H(26) = 26 \bmod 4 = 2$$

$$H(57) = 57 \bmod 4 = 1$$

$$H(81) = 81 \bmod 4 = 1$$

$$H(70) = 70 \bmod 4 = 2$$

$$H(64) = 64 \bmod 4 = 0$$

$$n = 4 \text{ y } r = 3$$

	0	1	2	3
1	96	57	26	115
2	48	81	70	79
3	64			35

$$D.O. = 10/12 = 83.33\% \text{ (Expansión parcial)}$$

$$H(115) = 115 \bmod 6 = 1$$

$$H(96) = 96 \bmod 6 = 0$$

$$H(48) = 48 \bmod 6 = 0$$

$$H(79) = 79 \bmod 6 = 1$$

$$H(35) = 35 \bmod 6 = 5$$

$$H(26) = 26 \bmod 6 = 2$$

$$H(57) = 57 \bmod 6 = 3$$

$$H(81) = 81 \bmod 6 = 3$$

$$H(70) = 70 \bmod 6 = 4$$

$$H(64) = 64 \bmod 6 = 4$$

$$H(107) = 107 \bmod 6 = 5$$

$$H(45) = 45 \bmod 6 = 3$$

$$H(62) = 62 \bmod 6 = 2$$

$$H(98) = 98 \bmod 6 = 2$$

$$H(33) = 33 \bmod 6 = 3$$

$n = 6$ y $r = 3$

	0	1	2	3	4	5
1	96	115	26	57	70	35
2	48	79	62	81	64	107
3			98	45		
Col				33		

D.O. = $15/18 = 83.33\%$ (Expansión parcial)

$H(115) = 115 \bmod 8 = 3$
 $H(96) = 96 \bmod 8 = 0$
 $H(48) = 48 \bmod 8 = 0$
 $H(79) = 79 \bmod 8 = 7$
 $H(35) = 35 \bmod 8 = 3$
 $H(26) = 26 \bmod 8 = 2$
 $H(57) = 57 \bmod 8 = 1$
 $H(81) = 81 \bmod 8 = 1$
 $H(70) = 70 \bmod 8 = 6$
 $H(64) = 64 \bmod 8 = 0$
 $H(107) = 107 \bmod 8 = 3$
 $H(45) = 45 \bmod 8 = 5$
 $H(62) = 62 \bmod 8 = 6$
 $H(98) = 98 \bmod 8 = 2$
 $H(33) = 33 \bmod 8 = 1$
 $H(28) = 28 \bmod 8 = 4$
 $H(38) = 38 \bmod 8 = 6$

$n = 8$ y $r = 3$

Estructura final y solución:

	0	1	2	3	4	5	6	7
1	96	57	26	115	28	45	70	79
2	48	81	98	35			62	
3	64	33		107			38	

2. Utilizando la estructura final obtenida en las expansiones totales y parciales, con densidad de ocupación para reducir $D.O. < 125\%$ y la función Hash $H(k) = k \bmod n$ eliminar paso a paso las siguientes claves: 48, 35, 81, 70, 45, 33, 38, 115 utilizando:

c. Reducciones totales

$n = 8$ y $r = 3$

	0	1	2	3	4	5	6	7
1	96	57	26	115	28	45	70	79
2	48	81	98	35			62	
3	64	33		107			38	

$$D.O. = 10/8 = 125\%$$

$$D.O. = 9/8 = 112.5\% \text{ (Reducción total)}$$

$$\text{Claves por eliminar: } 17 - 9 = 8$$

	0	1	2	3	4	5	6	7
1	96	57	26	115	28	45	70	79
2	48	81	98	35			62	
3	64	33		107			38	

	0	1	2	3	4	5	6	7
1	96	57	26	107	28		62	79
2	64		98					
3								

$$H(96) = 96 \bmod 4 = 0$$

$$H(79) = 79 \bmod 4 = 3$$

$$H(26) = 26 \bmod 4 = 2$$

$$H(57) = 57 \bmod 4 = 1$$

$$H(64) = 64 \bmod 4 = 0$$

$$H(107) = 107 \bmod 4 = 3$$

$$H(62) = 62 \bmod 4 = 2$$

$$H(98) = 98 \bmod 4 = 2$$

$$H(28) = 28 \bmod 4 = 0$$

$$n = 4 \text{ y } r = 3$$

Estructura final y solución:

	0	1	2	3
1	96	57	26	79
2	64		62	107
3	28		98	

d. Reducciones parciales

$$n = 8 \text{ y } r = 3$$

	0	1	2	3	4	5	6	7
1	96	57	26	115	28	45	70	79
2	48	81	98	35			62	
3	64	33		107			38	

D.O. = $10/8 = 125\%$

D.O. = $9/8 = 112.5\%$ (Reducción parcial)

Claves por eliminar: $17 - 9 = 8$

	0	1	2	3	4	5	6	7
1	96	57	26	115	28	45	70	79
2	48	81	98	35			62	
3	64	33		107			38	

	0	1	2	3	4	5	6	7
1	96	57	26	107	28		62	79
2	64		98					
3								

$H(96) = 96 \bmod 6 = 0$

$H(79) = 79 \bmod 6 = 1$

$H(26) = 26 \bmod 6 = 2$

$H(57) = 57 \bmod 6 = 3$

$H(64) = 64 \bmod 6 = 4$

$H(107) = 107 \bmod 6 = 5$

$H(62) = 62 \bmod 6 = 2$

$H(98) = 98 \bmod 6 = 2$

$H(28) = 28 \bmod 6 = 4$

$n = 6$ y $r = 3$

Estructura final y solución:

	0	1	2	3	4	5
1	96	79	26	57	64	107
2			62		28	
3			98			